



AZURE MASTER ARCHITECT PROGRAM · AUGUST 2026

Unified Digital Citizen Services Platform.

One front door for 2.1 M Nordic citizens across DK · SE · NO

ARCHITECTURE, AI & AGENTIC SUBMISSION

2.1 M

citizens · DK · SE · NO

12

languages

47 → 1

portals unified

28 → 4 d

processing time

What this is, and what it isn't.



A target vision

The production-target architecture for a federated Nordic platform, end to end.



A working core

Deployed on a real Azure tenant with one command; the live demo runs on it.



Some blueprint

A few bricks are designed and registered, not yet switched on, each with a roadmap gate.

Every claim in this deck carries one honesty label:



Live: on the tenant today



Implemented: merged & smoke-tested



Scripted: installer phase, idempotent



Blueprint: designed, not yet live



Roadmap: external dependency

UDCSP is a production accelerator: a target vision, a working core you can run today, and a named path for the rest.

The problem.

2.1 M

citizens across DK SE NO

47

separate legacy portals

28 d

average time for a residency case

A citizen who moves from Copenhagen to Stockholm proves their identity again, re-sends the same papers, waits 28 days for a decision, and uses a portal that may not speak their language, or may not be accessible at all.

UDCSP unifies the identity and the experience, then connects to the national systems to share and check information, never to replace them.

GDPR

EU AI Act

eIDAS 2.0

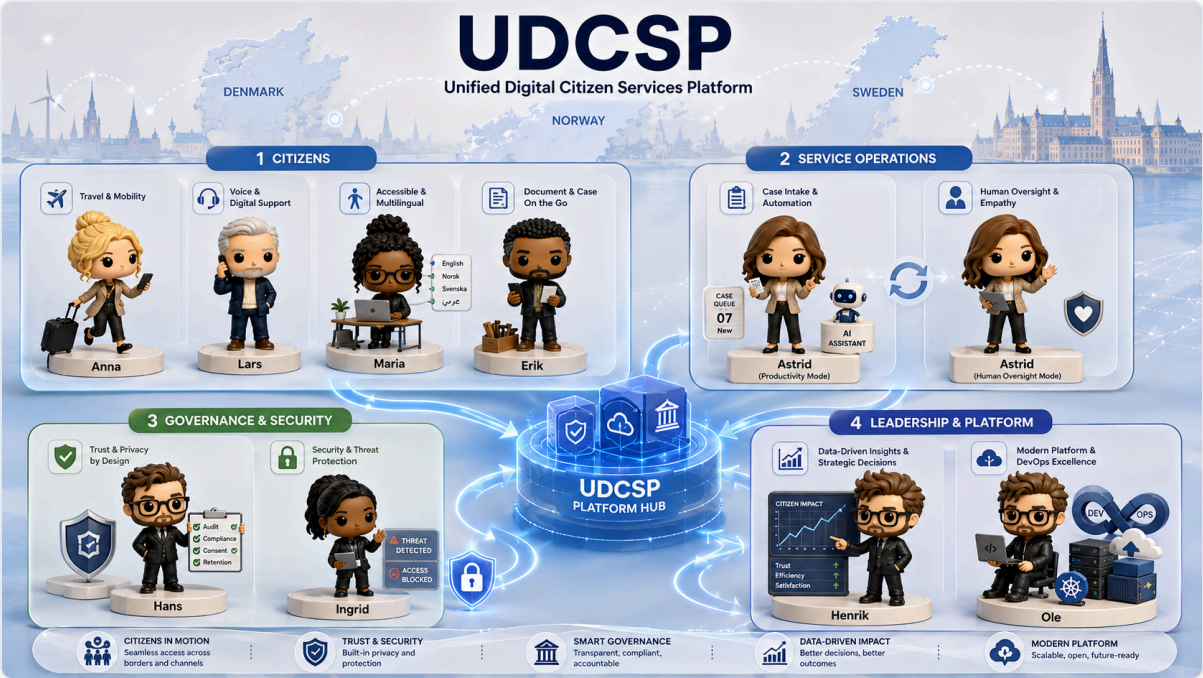
NIS2

WCAG 2.1 AA

ePrivacy



Who we build for.



CITIZENS

Anna · Lars · Maria · Erik

Anna moves country. Lars is blind and calls in. Maria uses a screen reader. Erik uploads from his phone.

OPERATIONS

Astrid, caseworker

The AI helps her work faster. She always makes the final decision.

GOVERNANCE

Hans · Ingrid

Hans protects privacy and consent. Ingrid watches for security threats.

LEADERSHIP

Henrik · Ole

Henrik needs clear data to decide. Ole runs the platform with one command.



The insight: unify the identity, connect the systems.

We did not rebuild the national portals. We unified the one thing that was missing: a single identity and a single experience, and connected it to the systems that already hold the records.

The citizen signs in once with their national eID. From there the platform shares and checks information with each authority, pre-filling the form, submitting it, following the status live. borger.dk, Skatteverket, NAV, Altinn and UDI stay the system of record and the decision-maker; we connect, we never replace.

WHY IDENTITY-FIRST, NOT ANOTHER PORTAL

- One identity, one design, one accessibility standard
- Information is shared and verified, not re-collected
- The same code on web, mobile and phone
- A new service is a new card on the same shelf, not a new website
- The authorities keep their data; we keep the experience

Sign in once. Share and check, don't re-collect. Follow the case live.

One front door: what the citizen lands on.

UDCSP Citizen Portal

Home
My cases
Consent
Accessibility
Compliance
Demos

EN · English

Hi Anna Denmark citizen
Sign out

NORDIC PUBLIC SERVICES · LIVE

Unified Digital Citizen Services

One trusted entry point for residents of Denmark, Sweden and Norway. Apply once, follow your case in real time, get help in your language — with caseworkers always in the loop.

Start an application
Track my cases

EU data residency · sovereign by design

12 languages · ICU MessageFormat, RTL ready

WCAG 2.1 AA · screen-reader & keyboard verified

EU AI Act · transparent, auditable, human override

What can we help you with today?

Pick a service and we'll guide you. Every form pre-fills what we already know.

Residency transfer

One guided intake. We pre-fill the country-specific registration steps and route you to the right authority — borger.dk / CPR · Skatteverket · Skatteetaten.

Get started →

Tax residency certificate

Request a tax residency certificate for DK, SE or NO. We pick the right form and pre-fill it from your eID data.

Get started →

Child & family benefit

Check eligibility and apply for child or family benefits. Upload payslip, lease or birth certificate — we extract the data and route to the competent authority.

Get started →

National Tax-administration

Reach Skatteetaten, Skattestyrelsen or Skatteverket by phone. Our AI voice assistant handles refunds, deductions and residency tax in your national language — with a human caseworker one keystroke away.

Get started →

AI consent active. Manage on [Consent & privacy](#).

Citizen assistant
Personalised

Hi Anna — ask me about your applications, your case status, or how to start a new service.

Send

Powered by **Foundry topic-router** via APIM. Every prompt and response is logged for audit (EU AI Act).

6

One identity, connected to every national system.

UDCSP Citizen Portal

[Home](#)
[My cases](#)
[Consent](#)
[Accessibility](#)
[Compliance](#)
[Demos](#)

EN - English

Hi Anna
Danmark citizen
[Sign out](#)

Get started →

Get started →

Get started →

Get started →

A unified platform across the Nordic public sector

UDCSP is a unified citizen platform that connects, in a single guided experience, the registers, eIDs and case systems of Denmark, Sweden and Norway. We don't replace national authorities — we collect your data once, validate the cross-border rules (Info Norden, Øresunddirekt, Grensetjänsten, EU Single Digital Gateway), pre-fill the right form, and submit it to the competent authority for your country.

UNIFIED PLATFORM
UDCSP
One guided intake

Denmark

borger.dk

Citizen portal

CPR

Population register

MitID

National eID

SKAT

Tax authority

Udbetaling DK

Family benefits

Sweden

Skatteverket

Tax & population

Försäkringskassan

Social insurance

BankID

eID

Freja+

eID

Norway

Skatteetaten

Tax & registry

NAV

Welfare & benefits

Altinn

Forms portal

UDI

Immigration

ID-porten

eID gateway

AI consent active. Manage on [Consent & privacy](#).

Citizen assistant

Personalised

Hi Anna — ask me about your applications, your case status, or how to start a new service.

[Send](#)

Powered by **Foundry topic-router** via APIM. Every prompt and response is logged for audit (EU AI Act).

How the platform works for you

Four simple promises behind every screen — your data stays at home, AI helps but never decides alone, the portal is usable by everyone, and you stay in charge of your information.

EU

Your data stays in your country

When you sign in as a Danish, Swedish or Norwegian resident, the portal talks only to that country's public-service systems. Nothing about you is copied or stored in another country.

AI helps, a person decides

AI can pre-read your documents, suggest the right form and give a first opinion on whether you're eligible — but a real caseworker reviews every decision and signs it off. You always see, in plain language, what the AI suggested and why.

Made for everyone

The portal is available in twelve languages, including Polish, Arabic, Sámi, Ukrainian and Finnish. You can use it entirely with a keyboard, with a screen reader, in high-contrast mode or with a dyslexia-friendly font.

You stay in charge

You decide what we may do with your data — share with another Nordic country, send you SMS reminders, forward a copy to a third party. You can change your mind at any time on the Consent page, and the change takes effect immediately.

How the front door routes.

One guided intake. The portal finds which authority owns the request and sends it there, using each citizen's national eID and the public cross-border rules.



- **Denmark:** borger.dk · MitID

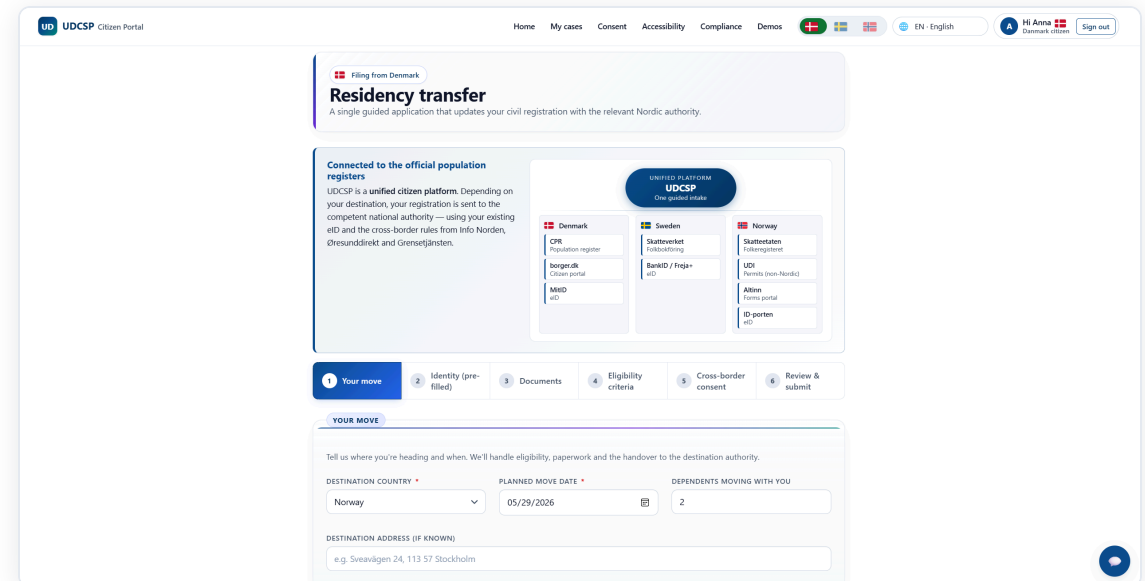


- **Sweden:** Skatteverket · BankID / Freja+



- **Norway:** UDI · Altinn · ID-porten

The rules come from Info Norden, Øresunddirekt and the EU Single Digital Gateway, following public policy, not logic we invented.



The screenshot displays the UDCSP Citizen Portal interface. At the top, there's a navigation bar with links like Home, My cases, Consent, Accessibility, Compliance, and Demos. A user profile for 'Hi Anne' is visible in the top right. The main content area is titled 'Residency transfer' and describes it as a single guided application for updating civil registration. Below this, a section titled 'Connected to the official population registers' explains the platform's role. A table lists the official registers for Denmark (CPR, borger.dk, MitID), Sweden (Skatteverket, BankID / Freja+, UDI), and Norway (Skatteetaten, UDI, Altinn, ID-porten). A progress bar shows six steps: 1. Your move (active), 2. Identity (pre-filled), 3. Documents, 4. Eligibility criteria, 5. Cross-border consent, and 6. Review & submit. The 'YOUR MOVE' section contains fields for 'DESTINATION COUNTRY' (set to Norway), 'PLANNED MOVE DATE' (05/29/2026), 'DEPENDENTS MOVING WITH YOU' (2), and 'DESTINATION ADDRESS (IF KNOWN)' (e.g. Svanavägen 24, 113 57 Stockholm).

What we ship.

CHANNEL

 **Web · Mobile · Voice**

One web app · 12 languages · accessibility-tested · one free phone number per country

AI BRAIN

 **One brain · 7 agents**

One orchestrator sends work to six specialists on Azure AI Foundry

OUTCOME

 **28 days → 4**

Cross-border residency in four days · 47 portals become one · 12 languages

SOVEREIGN

 **3 zones**

One AI hub and one data zone per country · no data crosses a border uninvited

UDCSP is a unified citizen platform, not a replacement. The national authorities still make the decision.

Sign in once: the rest is pre-filled.

UDCSP Citizen Portal

Home
My cases
Consent
Accessibility
Compliance
Demos

EN · English

Hi Anna Denmark citizen
Sign out

1 Your move
2 Identity (pre-filled)
3 Documents
4 Eligibility criteria
5 Cross-border consent
6 Review & submit

DOCUMENTS

Drop your employment contract — the **Document Extractor** agent (Foundry + AI Document Intelligence) reads employer, role, salary and start date. The file is stored in your country's lake (DK → udcspdkprod1lake) and never leaves the sovereign zone.

Upload employment contract (PDF / image)

employment_contract_anna_jensen.pdf · 5 KB

Preview
Open in new tab

EMPLOYMENT CONTRACT

Fictional sample document generated for demonstration purposes

This Employment Contract is made on **14 May 2026** between the Employer and the Employee named t
personal details, company details, dates, addresses, and compensation figures are fictional.

EMPLOYER	EMPLOYEE
Company Name: Northbridge Consulting Ltd.	Full Name: Anna Jensen

Extracted from your document — please confirm:

- **employeeName:** Anna Jensen
- **employer:** Nordic Retail A/S
- **employerCvr:** 12345678
- **period:** 2025-10
- **monthlyIncomeGross:** 36000

10

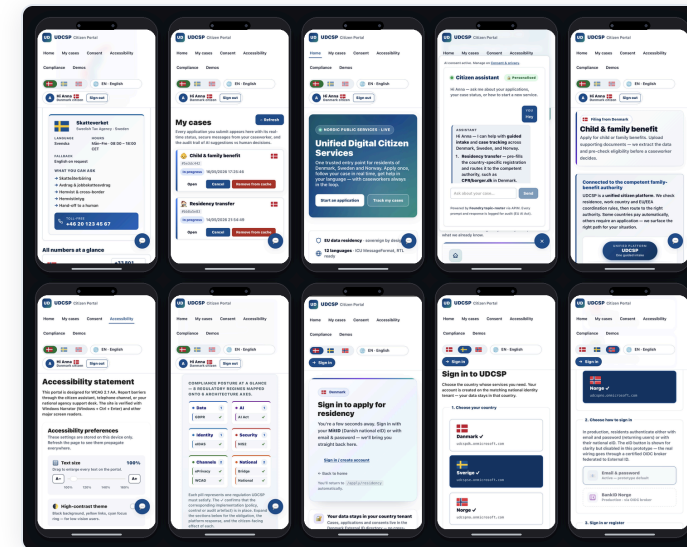
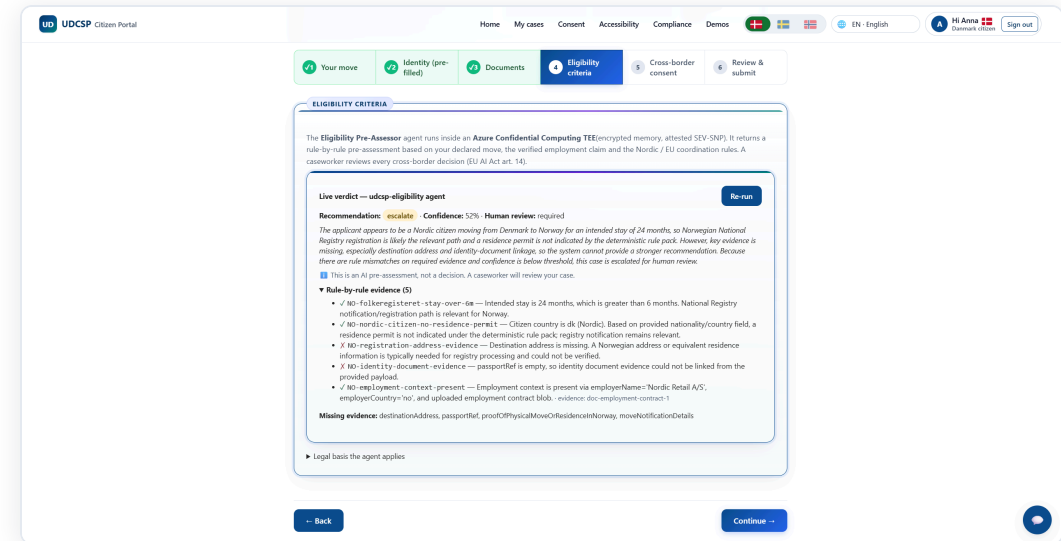
The platform you actually see.

One web app · 12 languages · 3 channels · always accessible.

The same React and TypeScript code runs on phone, tablet and laptop, and adapts to each screen.

The chat assistant stays in the corner. The accessibility menu offers slow speech, high contrast and reduced motion.

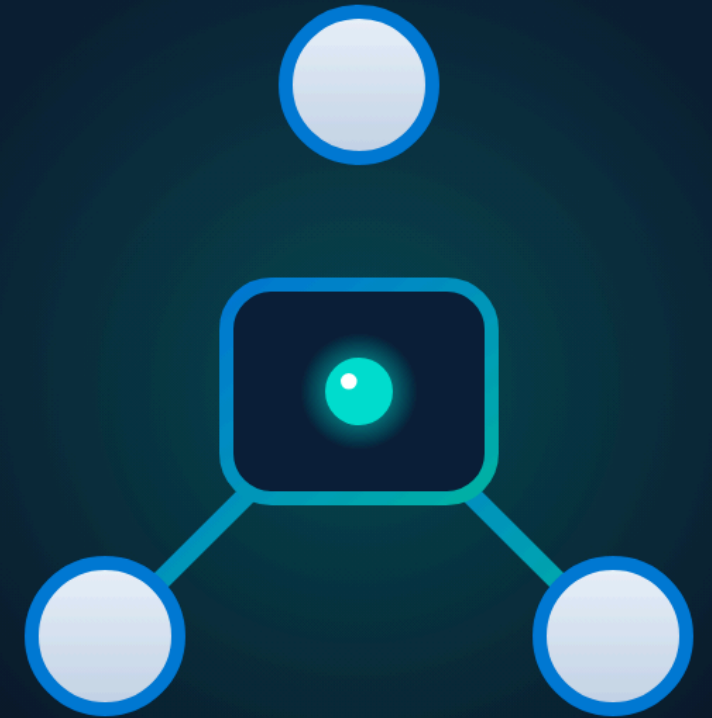
On every *Apply* page, the citizen sees the AI result (confidence, evidence, missing documents) before they consent.



Architecture.

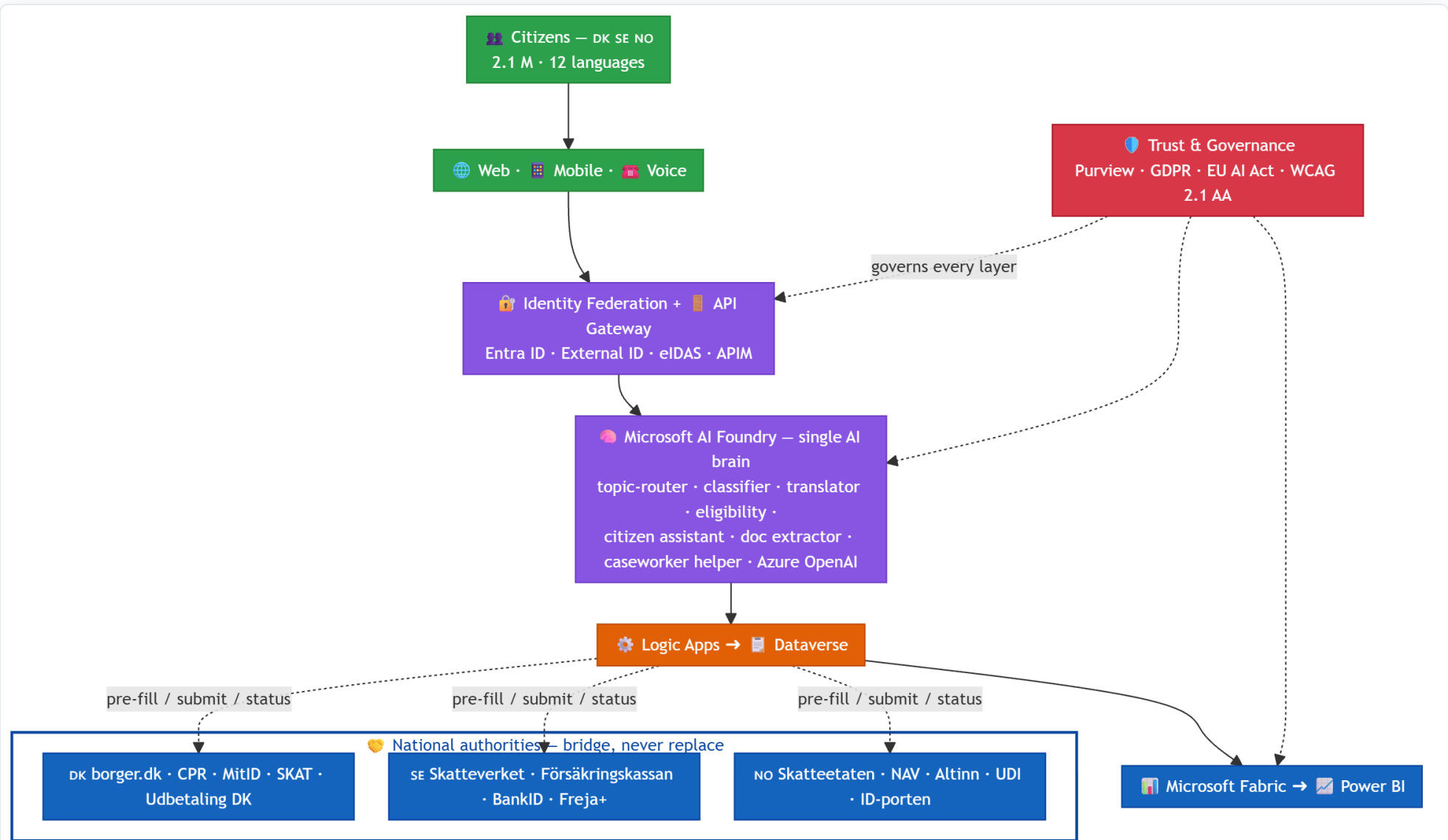
How we built it: three sovereign zones, one private path from the portal to the case

01





High-level architecture.



One country, one hub: three times.

A HUB PER COUNTRY

Each country has its own Azure region, private network, identity system, and AI hub on Azure AI Foundry.



- **Denmark:** North Europe region



- **Sweden:** Sweden Central region



- **Norway:** Norway East region

No AI call ever crosses a border.

THE RIGHT MODEL FOR EACH JOB

- **Real-time speech** model: the voice channel
- **Strong reasoning** model: the hard decisions
- **Fast** model: simple routing and sorting

The model name comes from one parameter file, so a region change is a one-line edit.

One honest exception. Real-time speech is not available yet in Norway, so Norwegian calls use the Sweden hub, inside the EU Data Boundary. The audio still stays in Norway. When the model arrives, one parameter brings it home.



Privatising every zone: from the portal to the case.

One public door per country. Everything behind it is private, from the citizen's first click to the caseworker's queue.

01 One public front door

Azure Front Door and a web firewall. Nothing else is public.

02 A private API gateway

Azure API Management checks the token, the only way in.

03 Backends on private endpoints

Workflows, agents, storage, database, vault: off the internet.

Private DNS per country.

04 The case lands in Dataverse

One case environment per country, in-region; Power Apps over Dataverse today, D365 Customer Service when licences land.

Writes go through workflows, reads through the gateway.

05 Analytics copies only numbers

Dataverse sends metrics to Microsoft Fabric. The raw rows stay in the country.

THE ONLY OTHER PUBLIC ADDRESS

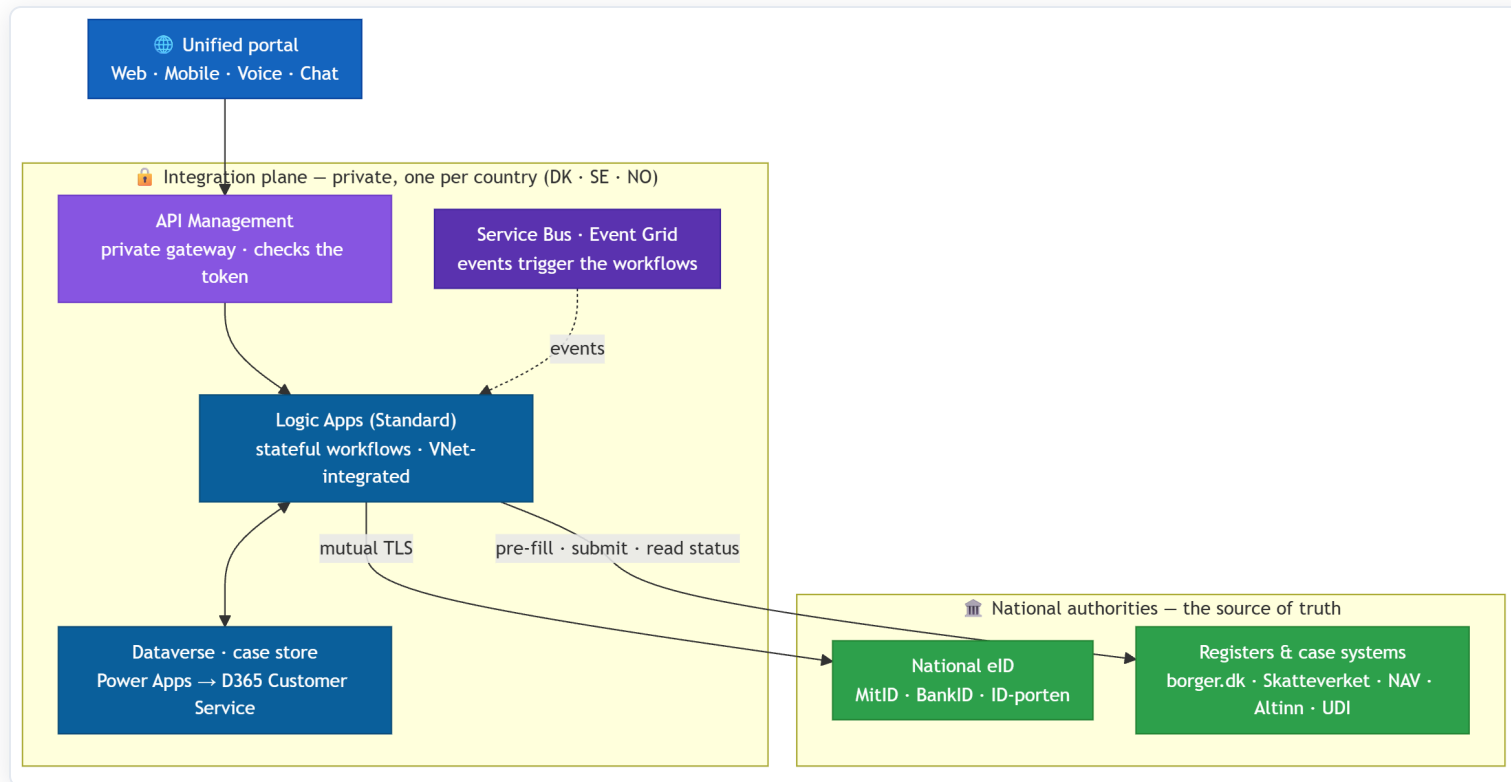
Azure Bastion is the single admin door per country. All outbound traffic goes through Azure Firewall, with allow-lists and TLS inspection.

ENCRYPTED EVERYWHERE

- Customer-managed keys, one set per country
- Two-way TLS to the national authorities
- Field-level encryption for national ID numbers

The integration bridge.

One private integration plane per country. The portal never calls a ministry directly; Logic Apps do, on its behalf.

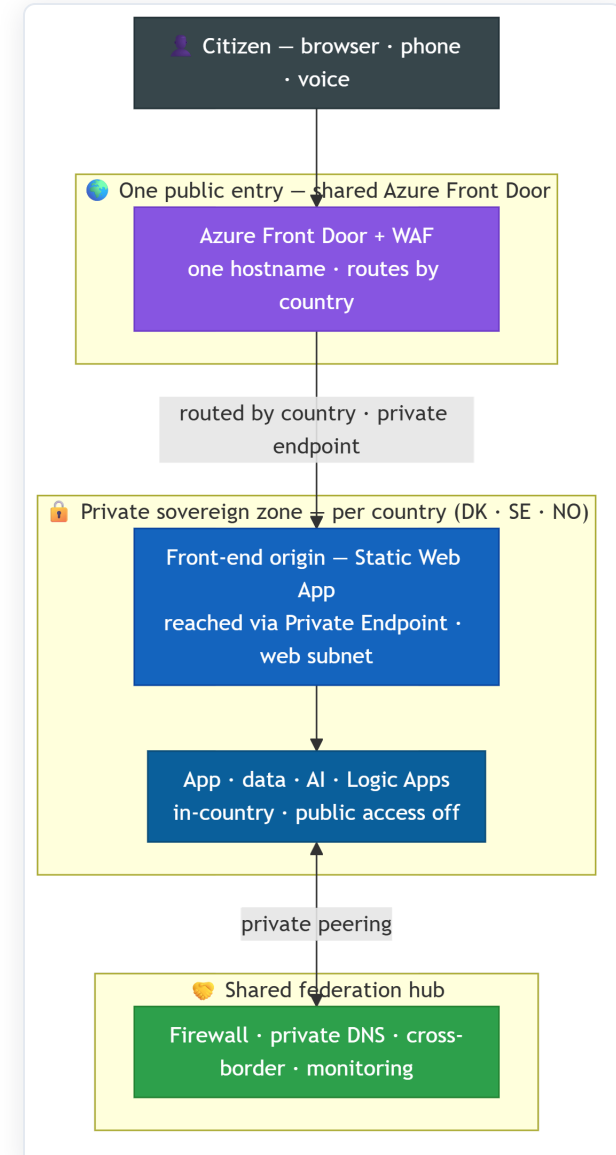


Logic Apps are the bridge: stateful workflows that pre-fill the national form, submit it, and read the status back, writing each case into Dataverse and reaching authorities over mutual TLS. In production they're VNet-integrated.

How the network fits together, simply.

One entry, one experience: private and sovereign underneath.

- **One public entry:** a single Azure Front Door (web firewall) and one UI codebase; the citizen always sees the same portal.
- **Sovereign by design:** each country has its *own* private front-end origin : a Static Web App reached through a private endpoint in that country's **web** subnet, with no public origin.
- That **web** subnet in each spoke is the private door to the front-end, not a separate copy of the app.
- App, data, AI, Logic Apps are private and in-country too; a shared hub handles firewall, DNS, cross-border and monitoring.





Design patterns we use.

Agents-as-Tools

On a phone call, the speech model decides by itself when to call a function or pass to a human.

Gateway + Saga

One API gateway per country. The cross-border case is a 6-step workflow that rolls back if a step fails.

System of record: Dataverse

Dataverse holds each case, surfaced via Power Apps today, D365 Customer Service tomorrow. Writes through workflows, reads through the gateway with a cache.

Strangler fig

The caseworker app writes to a generic Dataverse table today, the canonical D365 Customer Service case entity tomorrow with one switch.

Sidecar

The voice service pairs a call handler with a real-time audio bridge.

Circuit breaker

A failing national endpoint switches to a human queue; citizens never see a timeout.

Scaling to thousands of services.

01 Orchestration lives in one place

Per country, the Logic Apps integration plane behind the single gateway, never from the browser.

02 A journey is a saga

It fans out to each service in parallel, waits for all, and rolls back cleanly if one step fails.

03 A new service is declarative

A workflow + one gateway operation + one catalog entry, no new infrastructure.

04 Still findable

One front door, one search → the Topic Router routes to the right service, on web · mobile · voice, 12 languages.

Add the 200th service the same way as the 2nd: one workflow, one gateway operation, one catalog entry. The platform grows by configuration, not rewrites.

ONE VISA REQUEST → MANY SERVICES



- **Identity:** civil registry + national eID



- **Housing:** municipal address registry



- **Income:** employer + benefits data



- **Tax:** national tax authority



- **Vehicle:** vehicle registration



- **Healthcare:** health enrolment



Meeting old systems where they are.

Anti-corruption layer

One thin adapter per backend.
The mess of each old system stays at the edge; the core model stays clean.

Identity resolution

Phone number · national ID · case number, each mapped to one canonical citizen. The backend never adapts to us.

Two-way transformation

REST, SOAP or flat file → our common shape and back.
Request transform at the gateway, mapping in the workflow, mutual TLS to partners.

Once-only + catalog

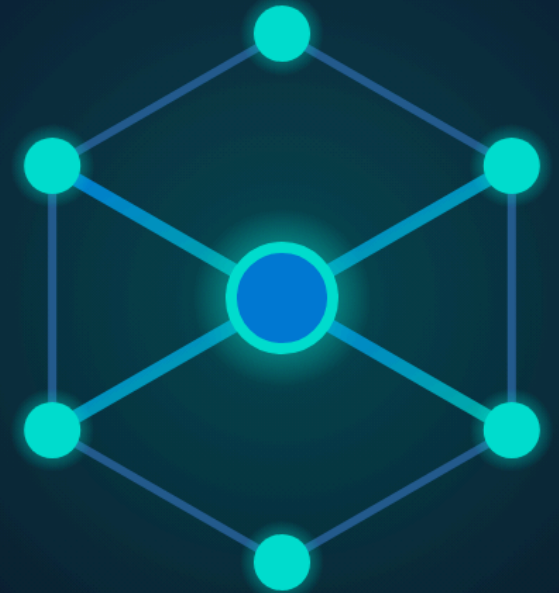
EU Once-Only evidence exchange means we never ask twice. Purview catalogs every field and its lineage.

We adapt to the backend; the backend never adapts to us. A new connector is configuration and an adapter, not a platform change.

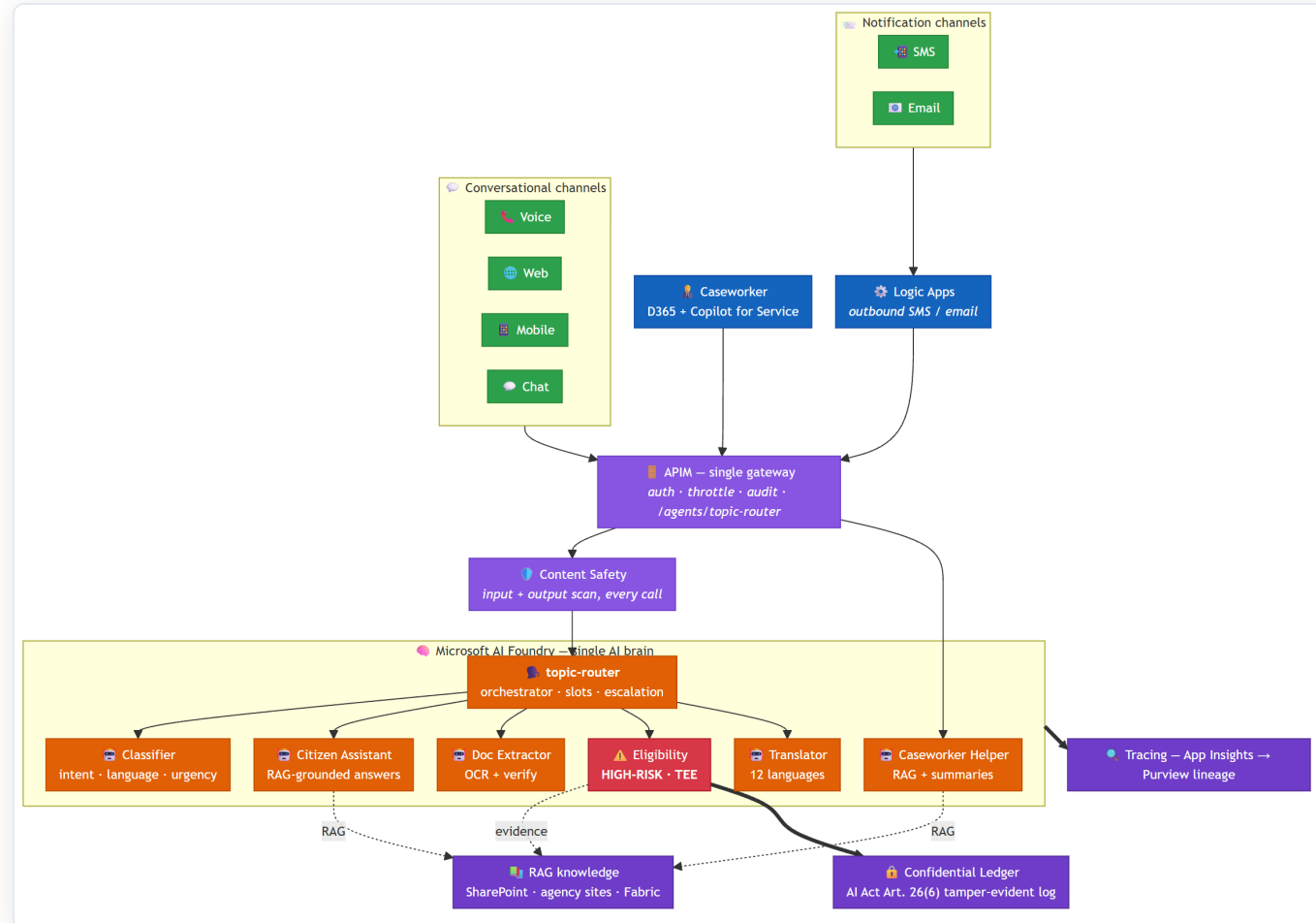
The AI Brain.

One brain on Azure AI Foundry · one orchestrator · six specialists · safe by design

02



The AI Brain: every channel, one gateway.



One brain, not seven chatbots: every model call goes through Azure AI Foundry, routed by the Topic Router.

How a citizen's question flows through the brain.

01 The channel asks the brain

Web, mobile and phone reach the Topic Router through the one private gateway, never a model directly.

02 The router understands and remembers

It finds the intent and the language across 12 languages, and keeps the short conversation for 24 hours.

03 It passes the task to a specialist

Classifier, Translator, Document Reader, Citizen Assistant, Caseworker Helper, or the high-risk Eligibility agent.

04 The specialist does one job well

The Assistant answers only from cited sources; the Reader extracts a payslip; Eligibility proposes a result with evidence.

05 The answer returns, traced and sourced

Every reply carries a trace number and its sources. On voice, the model decides when to call a tool or a human.

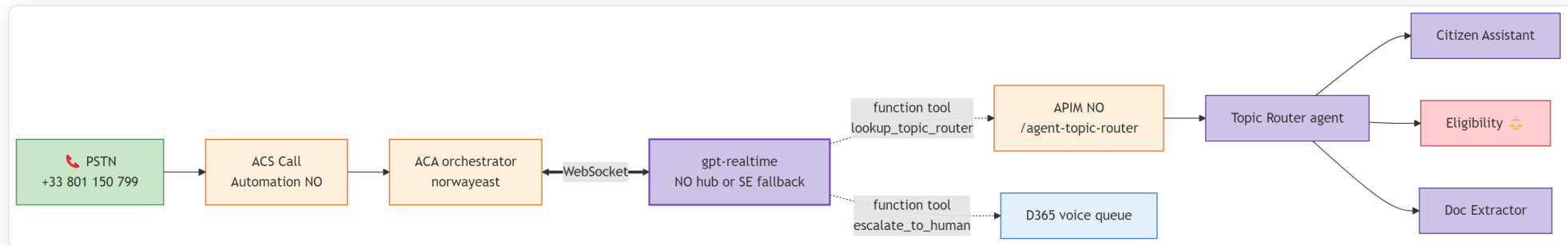
| Six specialist agents, two function tools, one orchestrator: seven agents in all, always under one human caseworker.

Agent catalogue.

Agent	Purpose	Model	EU AI Act class
Topic Router	Orchestration · 12 languages · slot-filling	Fast	Limited
Request Classifier	Intent · agency · language · urgency	Fast	Limited
Translator	12 languages · keeps admin terms exact	Strong + Translator	Limited
Eligibility Pre-Assessor 	Result + evidence · runs in a protected enclave · ledger-logged	Strong + rules	High (public service)
Citizen Assistant	Answers only from the cited public knowledge base	Strong, grounded	Limited
Document Reader	Passport / payslip / lease extraction	Fast + Doc Intelligence	Limited
Caseworker Helper	Summarise · draft replies · suggest next action	Strong, grounded	Limited

No agent makes a final decision. Every result is a *proposal* to a human caseworker, who confirms, changes or rejects it.

Voice channel: the model uses tools on its own.



Citizen dials → the call lands on Azure Communication Services → a small service opens a two-way audio stream to the speech model. The model decides on its own to call `lookup_topic_router` or `escalate_to_human` (a warm transfer). This is the Microsoft Agent Framework *Agents-as-Tools* pattern, on a real phone call.



Eligibility model lifecycle.

01 Champion-challenger

A new version runs in shadow on 5% of traffic for one week, then we evaluate it.

02 Parity gate per language

A gold test runs in all 12 languages. If one language is too weak, the release is blocked.

03 Drift detection

A daily test on inputs and outputs. Drift opens a security incident and forces a new evaluation.

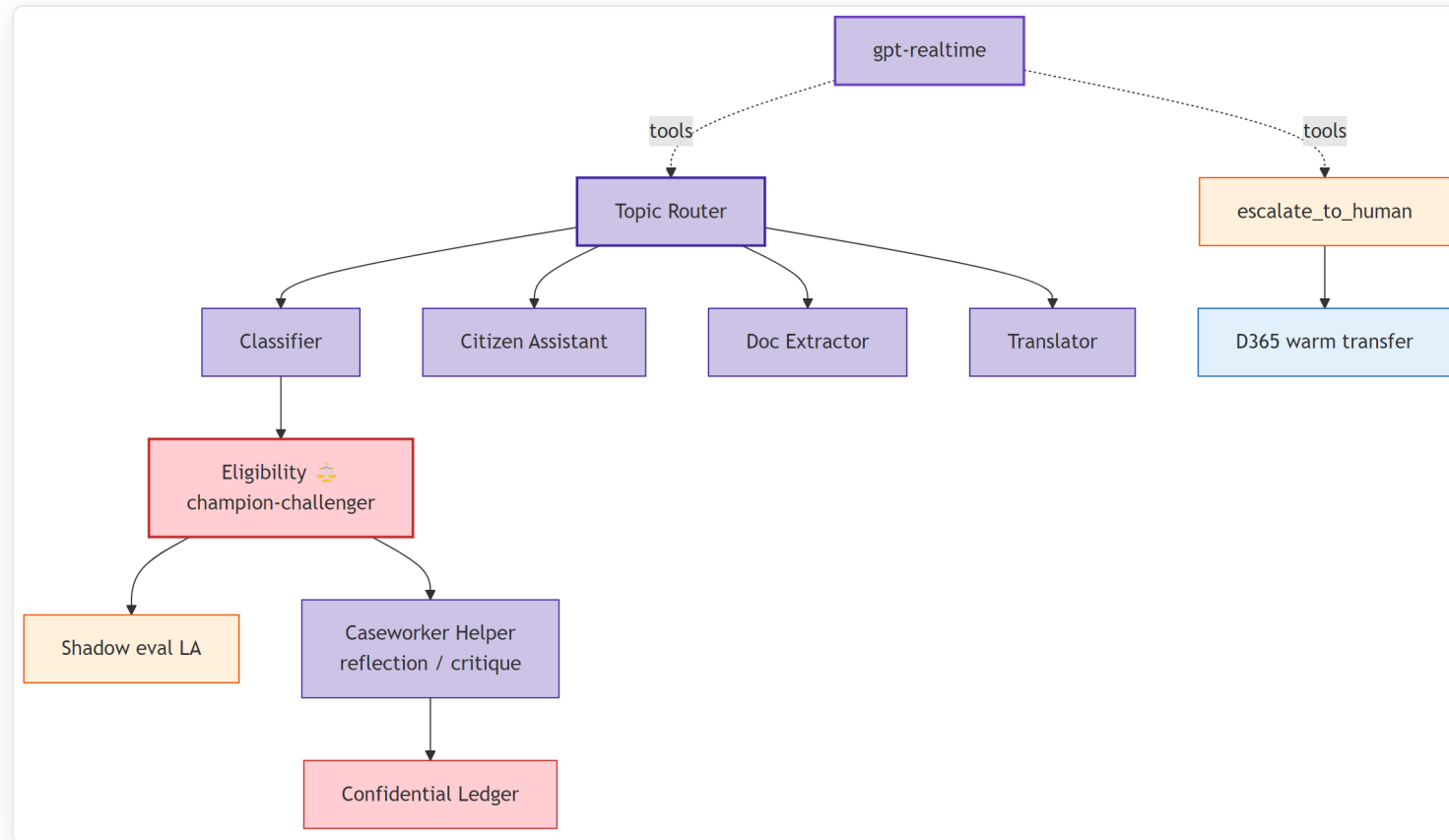
04 Bias monitoring

Fairness tests across age, location and channel for 30 days. A problem routes cases to a senior reviewer.

05 Rollback in seconds

Versions never change. Switching the alias restores the previous one at once, with an audit record.

Multi-agent coordination: beyond a single chatbot.



Handoff (router → six experts) · Reflection (the helper reviews the eligibility result) · State graph (the case is a 6-state workflow that can undo a step) · Function tool and warm transfer on voice · Shadow / canary for every model change.



Trust built into the brain.

Content Safety

A jailbreak and prompt-injection detector runs on every turn; a block raises a security event.

Evaluations

Quality is tested on every change: language parity, grounding and safety.

AI Act register

Each agent is declared with its purpose, risk level and limits, in a versioned file.

Confidential Ledger

Each high-risk result is hashed into a log that cannot be changed.

Human-in-the-loop

No agent decides. A caseworker confirms, changes or rejects every result.

Transparency

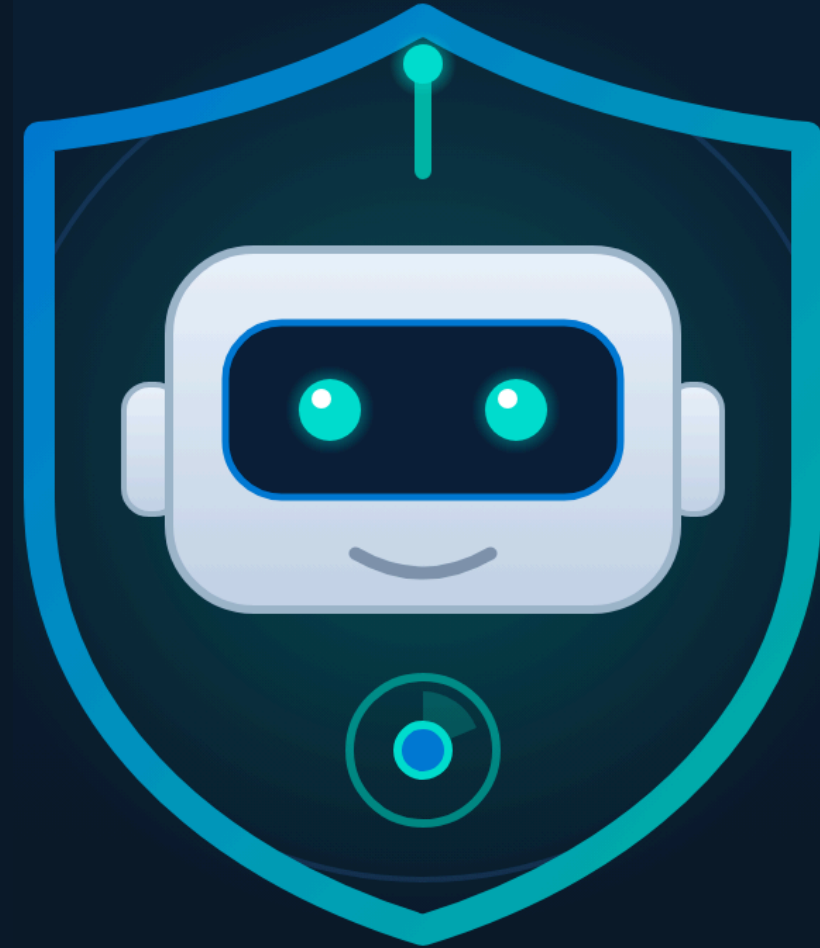
The citizen is always told AI is used: a chat badge and a spoken message in 12 languages.

The Eligibility agent is the only high-risk part. It runs inside a protected enclave (its data is encrypted in memory, even from an administrator), and it never decides alone.

UDCSP Guardian.

The next era: proactive, autonomous, human-supervised AI

03





From reactive to proactive.

20 to 60%

of eligible citizens never claim the benefits they are entitled to (OECD).
Not fraud, not choice: they do not know.

Today UDCSP is reactive: the citizen must know they qualify, find the portal, and apply. Guardian is the step after the EU once-only principle: no-stop-shop.

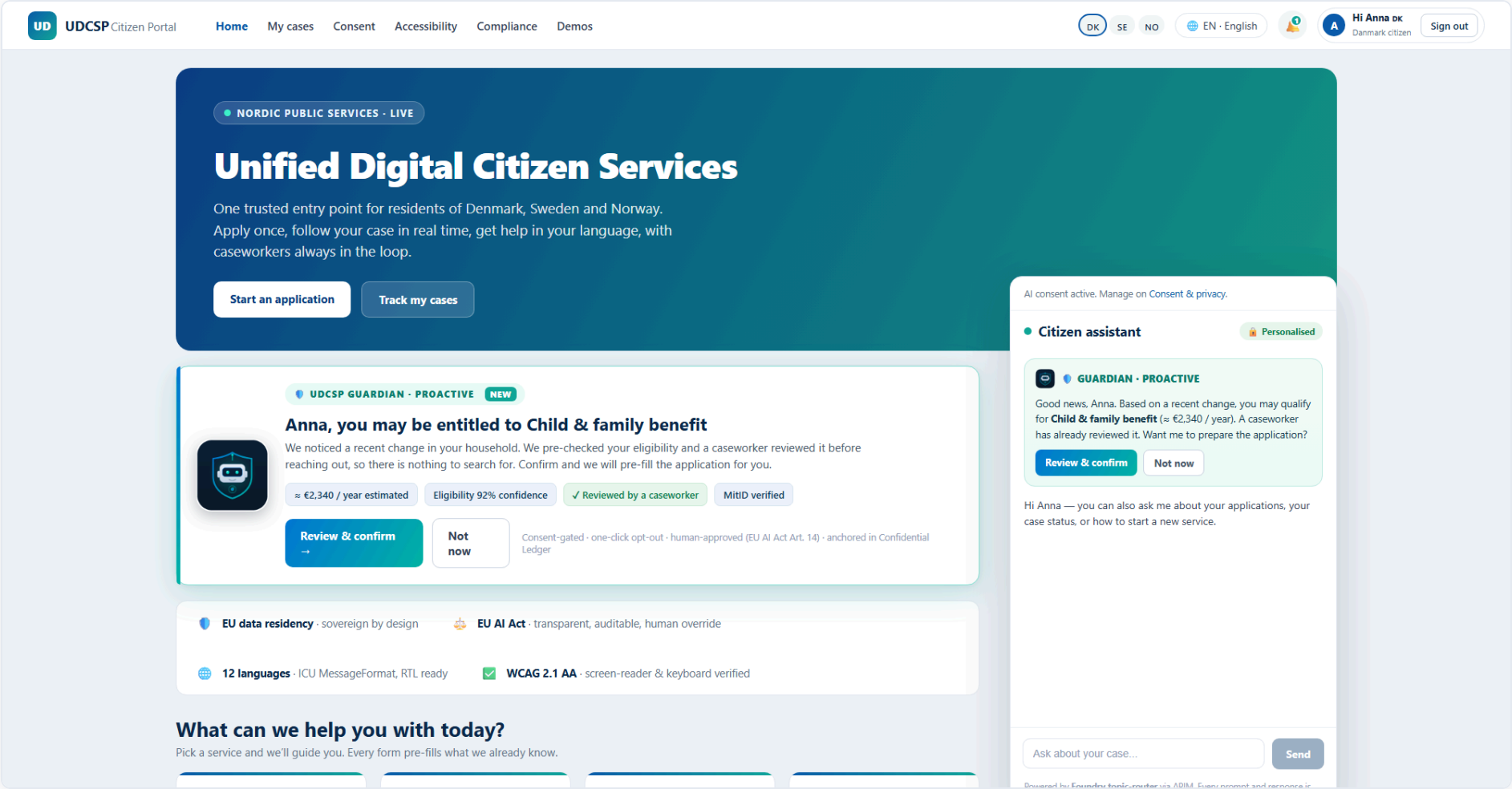
WHAT CHANGES

- The state detects the life event, not the citizen
- It reaches out first, in the citizen's language
- A human caseworker approves every message before it is sent
- Every step is consent-gated and ledger-anchored

Guardian turns a faster front door into a state that reaches out to the people it is meant to serve.



How Guardian appears to Anna.



| A proactive, caseworker-approved entitlement, surfaced right in the portal Anna already uses.
 August 2026

How Guardian works: one autonomous loop.

01 Event Scanner detects a life event

An autonomous planner scans in-country data: a birth, a move, age 67, an income drop.

02 Eligibility runs in shadow

The existing high-risk agent assesses silently, rule by rule, with no application attached.

03 Caseworker Helper drafts the outreach

A short, cited message in the citizen's language, from the next-best-action catalogue.

04 Critic agent reviews it **reflection**

A new agent checks legal basis, tone and false positives, and can send it back.

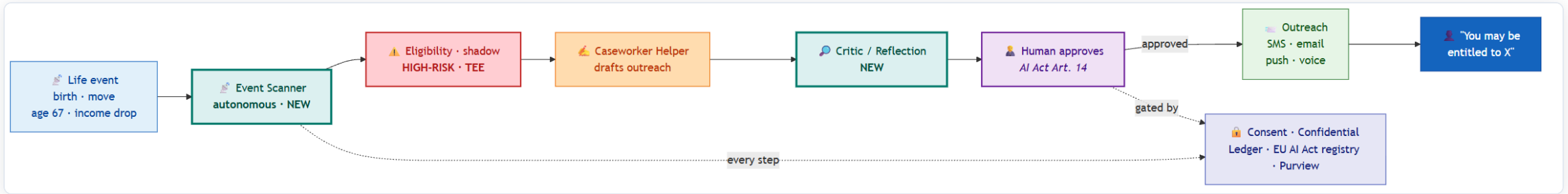
05 A human approves, adjusts or rejects **AI Act Art. 14**

The loop cannot advance without a caseworker. Nothing autonomous past this gate.

06 Outreach goes out, every step anchored

SMS, email, push or voice, consent-gated, hashed into Confidential Ledger.

The architecture: new engine, reused brain.



MOSTLY ASSEMBLY

- reused Eligibility (shadow), Caseworker Helper, Translator, the outreach channels, the ledger and registry
- new Event Scanner, Critic agent, one outreach workflow, one KPI tile

SOVEREIGN AND SAFE

- A Danish signal is assessed by the Danish brain, no border crossing without consent
- The high-risk lane is unchanged: sealed enclave, ledger-anchored, never decides

Two new agents wrapped around a platform that is already live, private and governed.

Responsible autonomy, and the number that matters.

RESPONSIBLE BY DESIGN

- **GDPR Art. 22** · never an automated decision with legal effect; a human approves
- **EU AI Act Art. 14** · mandatory human oversight kept in the loop
- **Consent** · lawful basis required, one-click opt-out, checked before sending
- **Ledger** · signal, verdict, draft and disposition all anchored

€ recovered

the new executive KPI: unclaimed entitlements proactively delivered, by country and language

FROM EFFICIENCY TO EQUITY

- Reactive story: 28 days to 4
- Guardian story: money and rights reaching people who never knew

Proactive government is safe when autonomy stops at a human, the basis is lawful, the citizen can opt out, and every step is provable.

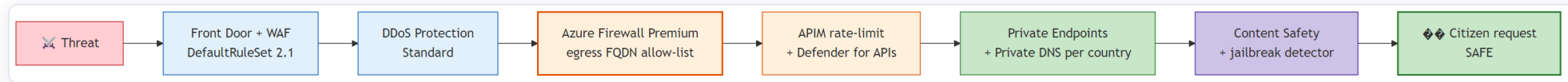
Security & compliance.

How we made it provable: eight EU laws mapped to controls in code

04



Defence in depth: many layers.



Edge: application

Azure Front Door and web firewall: common-attack rules, bot protection, rate limits

Edge: network

Standard DDoS protection on every network

Traffic out

Azure Firewall · destination allow-lists · TLS inspection



Identity & data sovereignty.

IDENTITY

- One citizen-identity system per country (Microsoft Entra External ID)
- National eID via a certified broker: MitID · BankID + Freja+ · ID-porten
- Support for the EU Digital Identity Wallet
- Admins get time-limited access only, through one jump-host

One encryption key set per country · modern TLS in transit · two-way TLS to partners · field-level encryption for national IDs · public network access turned off everywhere.

DATA: FIVE ZONES, ALL IN-COUNTRY

- **Live:** managed database and cache
- **Documents:** write-once, cannot be changed
- **Conversations:** kept for AI Act records
- **Knowledge:** search, locked per citizen and country
- **Analytics:** sovereign EU Fabric capacity

How we made it compliant: the story.

01 We started from the law, not the features

Eight bodies of EU law touch one interaction: a single voice call is data protection, accessibility, e-identity, cybersecurity and AI law, all at once.

02 Each obligation became a control in code

Every rule maps to a real platform mechanism, with an evidence file behind it.

03 We made the evidence provable

One trace number follows every interaction, from the browser to the model and back, kept for two years.

GDPR: data protection

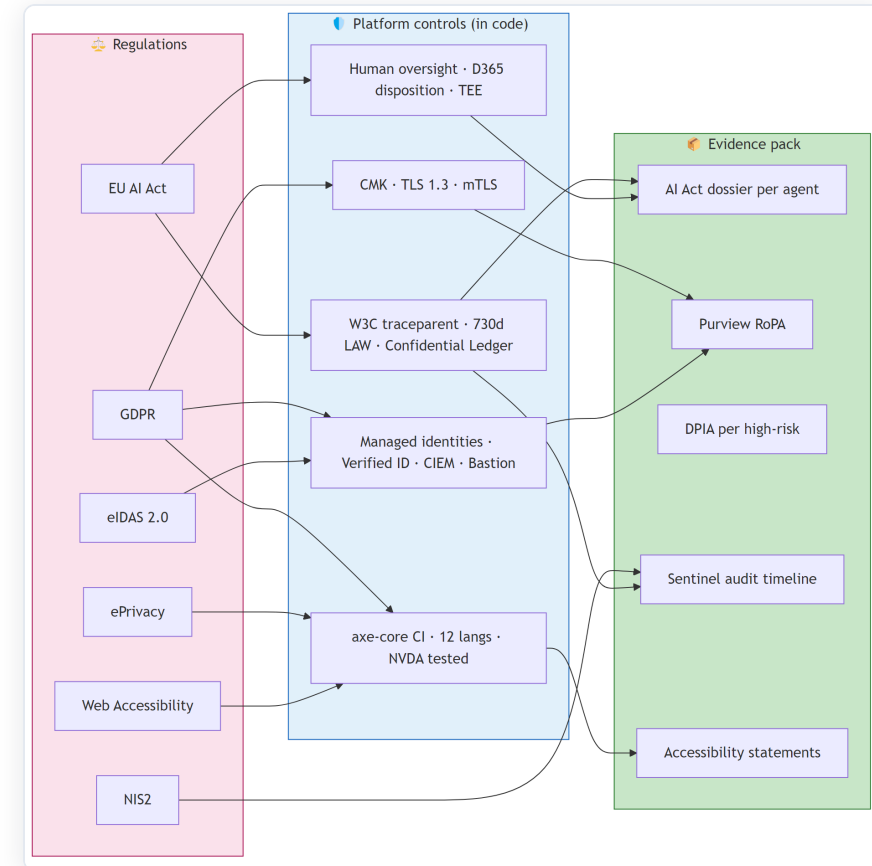
EU AI Act: high-risk AI

eIDAS: e-identity

NIS2: cybersecurity

ePrivacy: cookies

WCAG: accessibility





EU AI Act: article by article.

Article	Requirement	UDCSP delivery
Art. 12	Automatic records ≥ 6 months for high-risk	Model logs kept 730 days = 2× the minimum
Art. 13	Transparency for deployers	A register entry per agent · tests per language
Art. 14	Human oversight	A caseworker confirms every result · ledger-logged
Annex III §5(b)	High-risk = access to essential public services	The Eligibility agent is declared high-risk
Art. 50	The citizen must know AI is used	AI badge on chat · spoken message in 12 languages

The AI Act, GDPR, ePrivacy, eIDAS, NIS2 and accessibility law are each mapped to platform controls. The full mapping is in `docs/biz/datacompliance.md`.



Security in action: a prompt injection, stopped.

A message that tries to leak the hidden instructions, or change a result, is caught at three independent layers (three different Azure tools):

01 Azure API Management

The gateway sees the strange token rate and blocks the data leak.

02 Azure AI Content Safety

The jailbreak detector raises a security event and opens an incident.

03 Fixed eligibility rule

A deterministic plug-in rejects the request before any model runs.

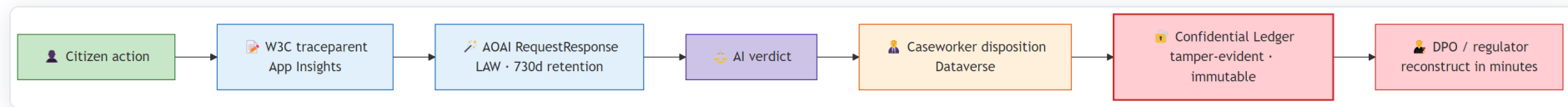
THE AUTOMATIC RESPONSE

- The session is isolated at once
- The citizen flow is brought back safely
- An audit pack is saved for review

0

citizen data exposed

Compliance in action: replay a decision, months later.



Hans, a data-protection officer, takes the citizen's trace number and rebuilds the whole decision: the model, the tokens, the result, the human choice, and the cryptographic ledger proof.

The AI Act minimum is six months. We keep two years.
From the request to a full audit pack: under 10 minutes.

Monitoring & FinOps.

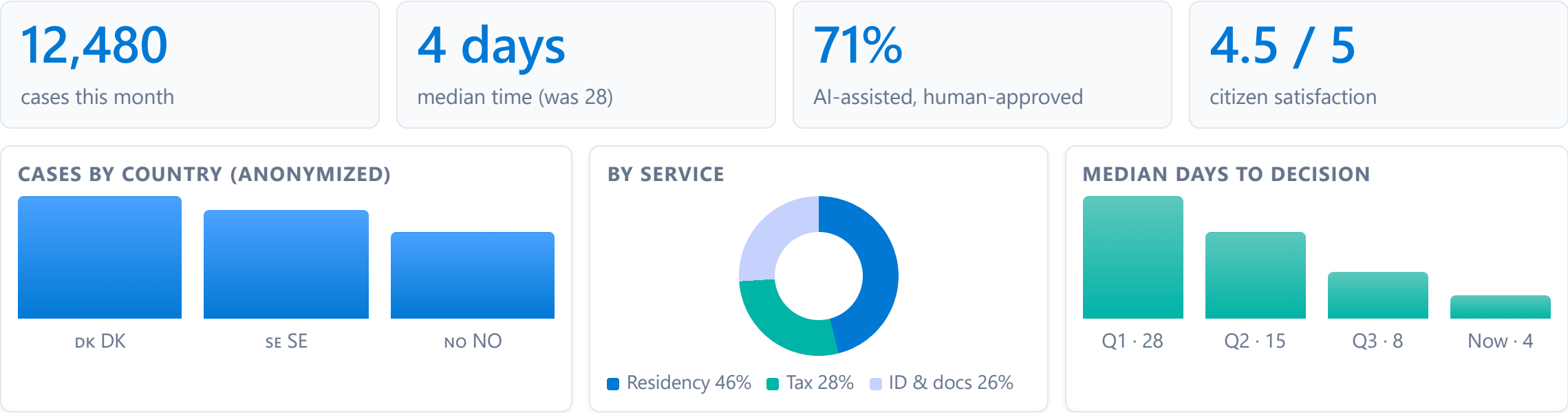
One anonymized central view · end-to-end tracing · cost per token





One central view: anonymized, in Fabric & Power BI.

Leadership needs the big picture. But raw citizen data never leaves its country. So only anonymized, aggregated numbers travel to a central view.



Only anonymized, aggregated numbers leave each country: never a name, never a case file. Built on Microsoft Fabric and Power BI · refreshed every hour.



FinOps: cost under control.

ALLOCATION & SHOWBACK

- Every resource tagged `country` , `workload` , `cost-center`
- Cost views per country and per workload
- Each ministry sees its own country's cost

TOKEN BUDGET & CAPACITY

- A monthly token budget per agent, written in code
- The build fails if the total is over the pool
- Reserved capacity for steady models · pay-as-you-go for peaks
- Alert on an unusual daily cost jump

| We treat AI tokens like any other budget line: planned, tagged, and checked at build time.

Cost & economics.

What it runs · what it costs per citizen · and how we keep it low





Three tiers, one platform

€60k

/ month · pilot · 30k citizens

€144k

/ month · regional · 300k

€5.28M

/ month · national · 22.3M ceiling

€2.84

per citizen / year at scale

Scale	Citizens	Active / mo	Peak sessions	Run-rate / mo	Per citizen / yr
Pilot · early band	30 000	12 000	≈ 240	≈ €60 k	≈ €24
Regional · early band	300 000	120 000	≈ 2 400	≈ €144 k	≈ €5.8
National · DK + SE + NO	22 300 000	8.92 M	≈ 178 000	≈ €5.28 M	≈ €2.84

The same platform sized for three population bands. National = real populations, full model in `docs/biz/cost.md`.



Same platform, three regions, and the region changes the price

Country	Pop.	Azure region	Index	€ / cit. / yr
 Denmark	6.0 M	North Europe	1.00	€2.69
 Sweden	10.7 M	Sweden Central	1.08	€2.83
 Norway	5.6 M	Norway East	1.20	€3.04
All	22.3 M	three regions	-	€2.84

Index = Azure-infrastructure price vs North Europe. Licences (D365, identity) are region-neutral.

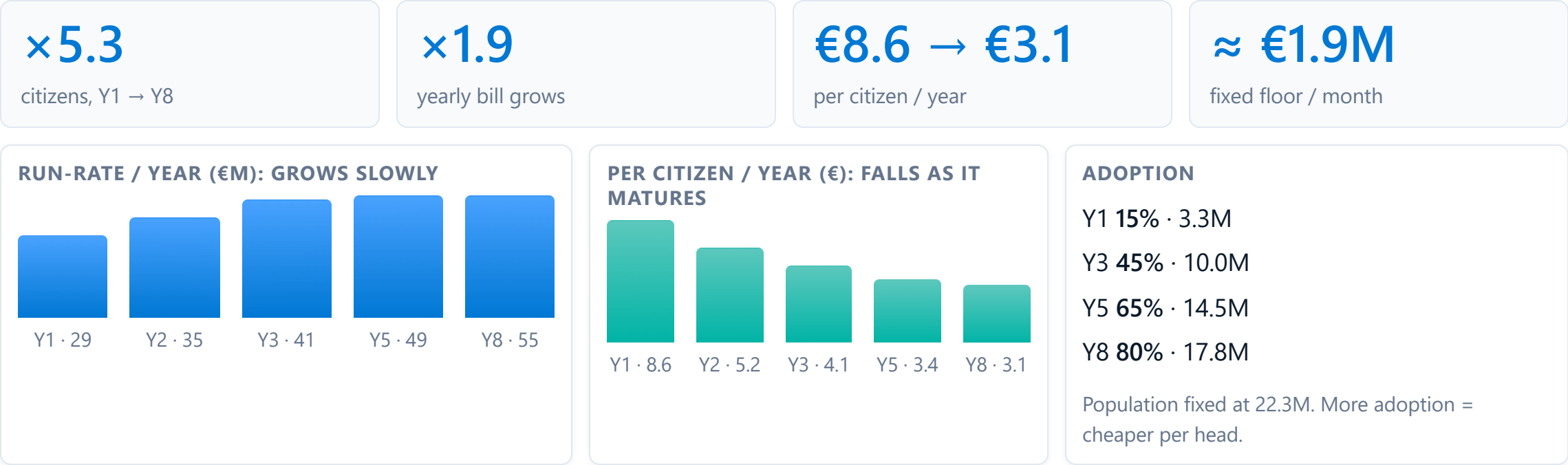
Per citizen / year: DK €2.69 · SE €2.83 · NO €3.04. The region, not the code, drives the gap.



Norway East premium + **smallest population** → highest per-citizen cost.
No country subsidises another.



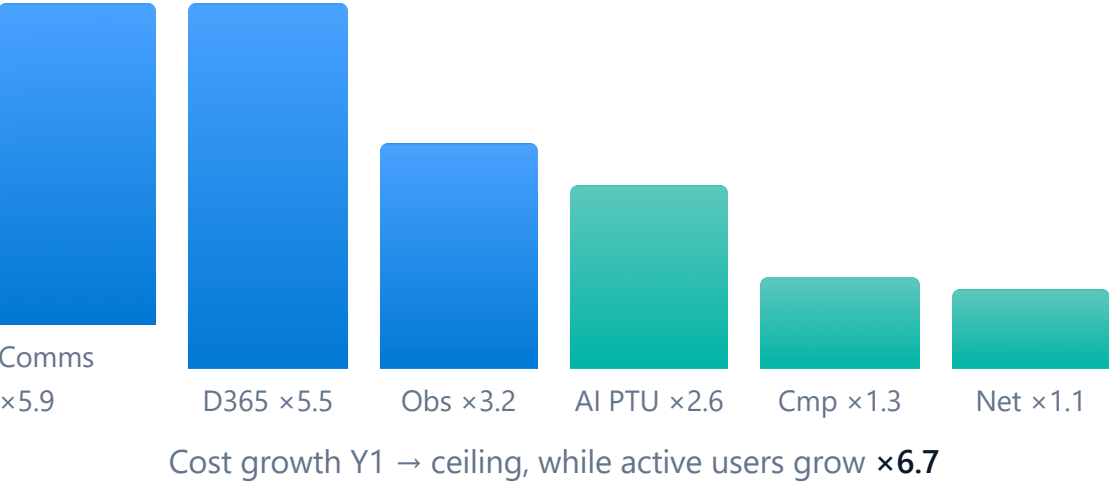
Cost over time: adoption ramps, so the bill ramps



Citizens ×5.3, bill only ×1.9: the fixed floor is paid once, then amortised. Per-citizen falls €8.6 → €3.1.



Not every service scales the same way



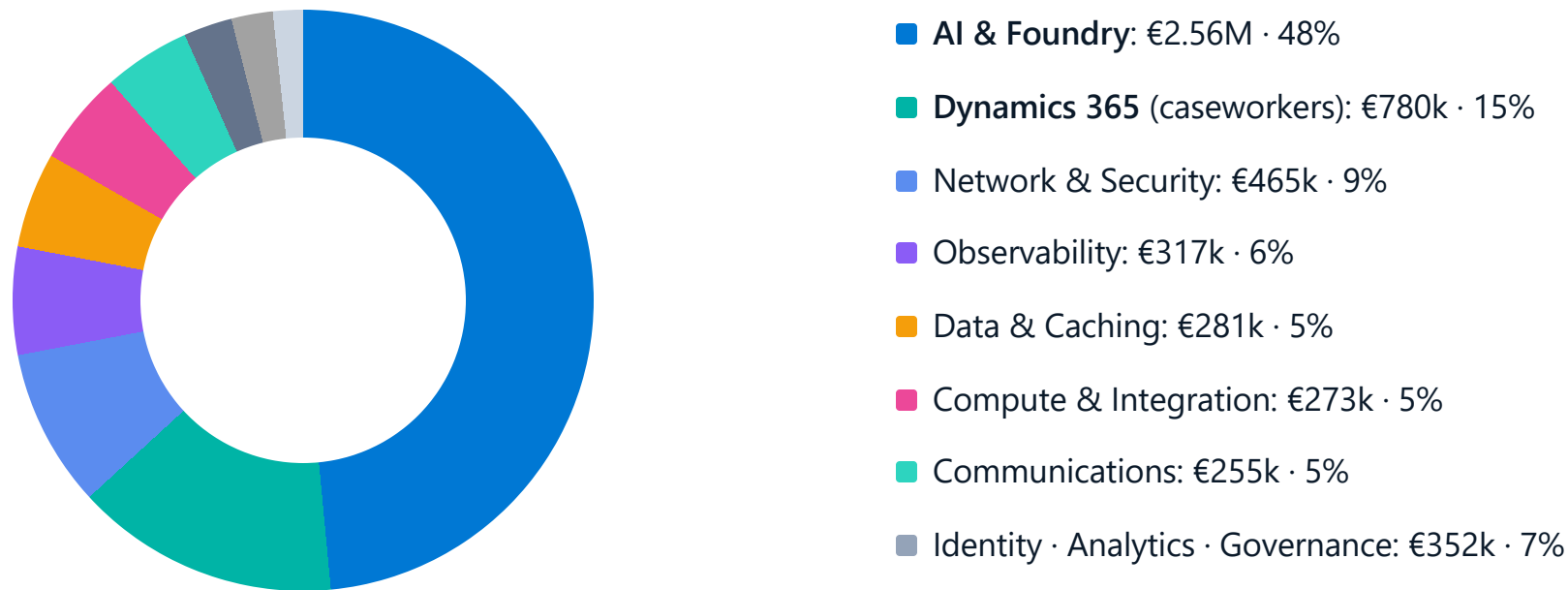
Service	Shape	Y1→ceiling
ACS SMS · voice · email	 metered	×5.9
Dynamics 365 caseworkers	 headcount	×5.5
AI PTU pools	 block	×2.6
APIM · Logic Apps · SKUs	 stepped	×1.3
Firewalls · Front Door · net	 flat	×1.1

APIM & Logic Apps jump in SKU steps, not per user. AI PTU is reserved to peak; the mini-first router keeps it flat.

Only ≈ €3.4M of the €5.28M ceiling scales with citizens; ≈ €1.9M is built once. +1 active citizen = €0.38 / month, €0 floor. Full breakdown in `docs/biz/cost.md` §7.



Where the money goes: €5.28M / month at the ceiling



Top two groups ≈ **63%** of €5.28M / month

Every slice traces to a deployed service; per-group arithmetic in `docs/biz/cost.md` §3-5 & §11.



Fixed floor vs variable: the cost flips with scale

Pilot · €60k / month, mostly fixed floor



National · €5.28M / month, mostly real usage



Driver · per active citizen / month	Value	What it hits
AI conversations (≈ 6 turns)	0.8	reserved AI capacity
SMS	0.6	Communications
Eligibility assessments	0.15	strong model + enclave
Escalation to a caseworker	≈ 4%	Dynamics 365 licences

Fixed = three sovereign zones · Variable tracks active citizens. Peak concurrency ≈ 2% of active users; that is all we reserve.



Keeping it low: levers already built in

LIST → COMMITTED



–30% to –40% off the AI & compute lines: same citizen experience.

Lever	What it does	Saving
Reservations & Savings Plans	1-3 yr commit on PTU, App Service, PostgreSQL, Fabric	–30 to –40%
Model routing (mini-first)	cheap mini routes; strong model only to reason	≈ 10× / token
PTU right-sized to peak	reserve to peak concurrency · pay-as-you-go spikes	no off-peak waste
Push & email before SMS	SMS is the most usage-sensitive line	can halve comms
Telemetry sampling	adaptive sampling · basic logs · cheap archive	–40 to –60% obs
Tagged showback	build fails if an agent goes over budget	stops drift early

List ≈ €63.4 M/yr national → ≈ €40-46 M committed. Full lever list in `docs/biz/cost.md` §9.

Performance & reliability.

Fast for the citizen, always available: measured targets, automatic failover, chaos drills

07





Performance & reliability.

99.9%

citizen web target, per country

≤ 2 s

voice answer, 95% of calls

≤ 1 h

recovery point: case data (15 min for drafts)

4 h

recovery time to the backup EU region

Active-passive per country

Azure Front Door sends traffic to the healthy region. Failover in under 5 minutes.

Backup & disaster recovery

Database, cache, storage and agent hosts. One recovery vault per country.

Chaos drills

We break things on purpose: region failover, network cut, database failover. Monthly in test, quarterly in production.

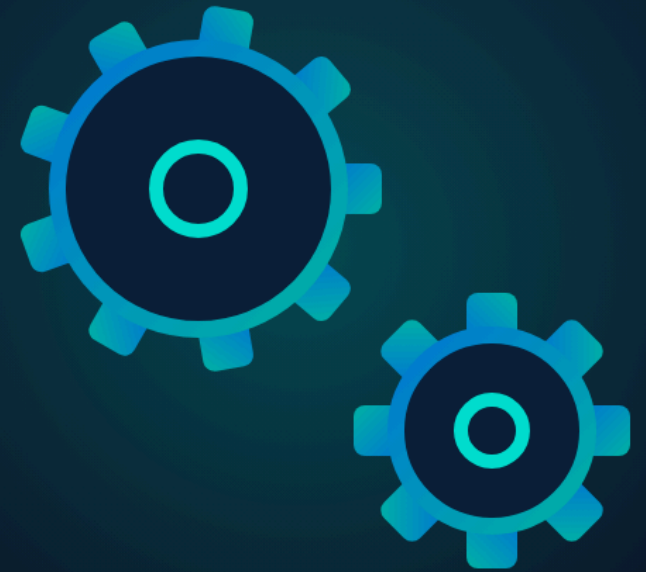
Autoscale & caching

Services scale out under load. Reads are cached at the gateway, so peaks stay fast.

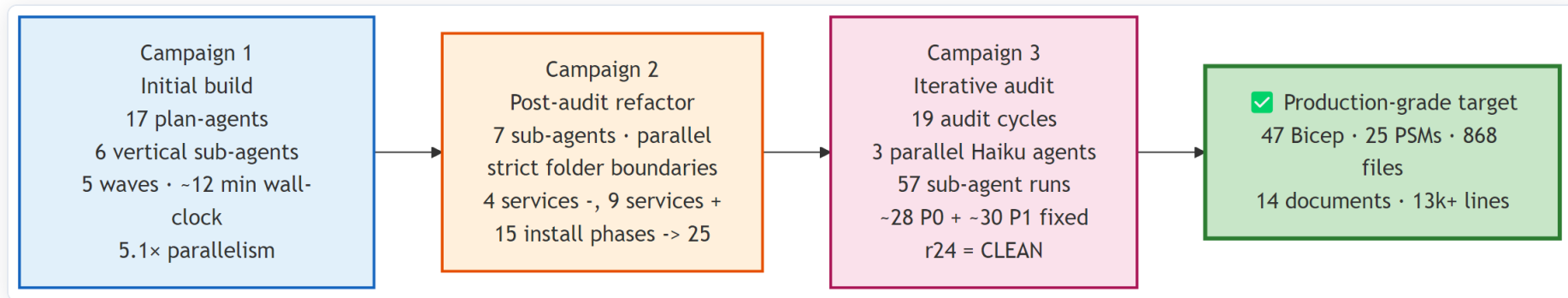
How we built it.

The platform itself was built by an agent swarm, in under 45 minutes of wall-clock time

08



UDCSP was built by an agent swarm.

**3**

multi-agent campaigns

~5×

faster than sequential

24

audit rounds to first clean

~1,000

files tracked in the repo today

Three campaigns (build, refactor, then repeated audits) produced the platform. Six sub-agents owned separate folders and ran in parallel. Each round produced fix commits until round 24 was the first fully clean one. The one-command installer is the executable summary of it all.

Everything is in the open: one repository.

THE STORY STARTS IN THE DOCS

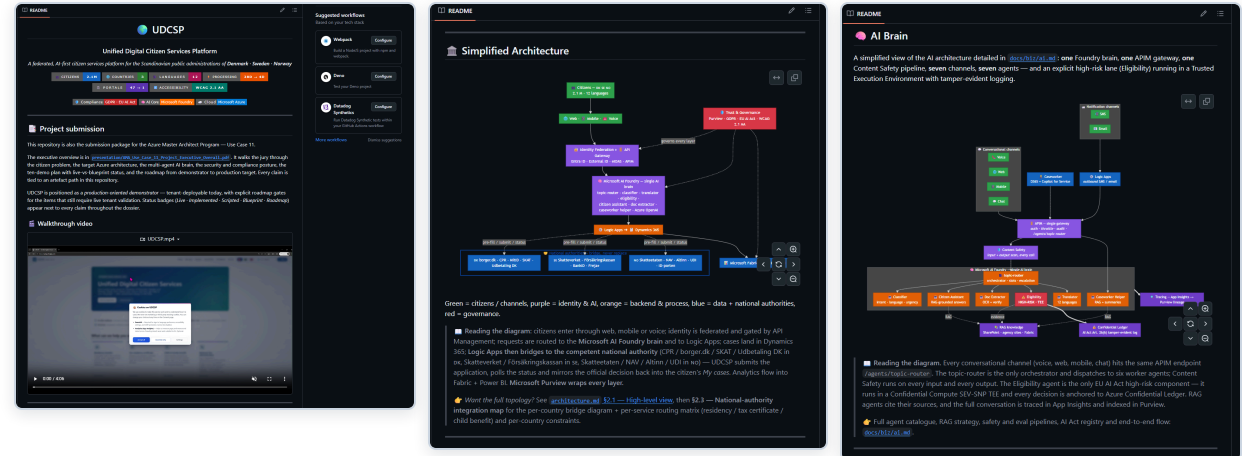
- `/docs/biz` : the business story: who we build for, every channel, data-and-compliance, demo scenarios, and an acceptance recipe for "done".
- `/docs/tech` : the engineering: architecture deep-dive, data model, network design, one-command install guide, disaster-recovery runbook.
- `/infra` : the Bicep that builds every zone
- `/scripts` : the installer.

SHORT LINK

aka.ms/UDCSP

GITHUB

github.com/fredgis/UDCSP



One idempotent deploy: provision.

One script, one click, and you can run it again safely.

- Checks prerequisites, signs in to Azure, registers the Entra ID identity app.
- Installs, builds, and runs the tests before anything is deployed.
- Provisions all the infrastructure as code via Bicep, through `azd`.
- **Idempotent:** a second run reuses what exists, never a duplicate.

```
Fabric Storyboard Copilot - One-Click Deploy

▶ Checking prerequisites
✓ Node.js v24.14.0
✓ npm v11.9.0
✓ Azure CLI v
✓ Azure Developer CLI (azd)

▶ Authenticating with Azure
✓ Signed in as
✓ Subscription:
Authenticating azd (fresh login)...
✓ azd authenticated

▶ Gathering configuration
Environment name (e.g., dev, staging, prod) [dev]:
✓ Environment: dev
Azure region [eastus2]:
✓ Location: eastus2
Deploy a NEW Azure OpenAI resource? (y/n) [n]:
Existing Azure OpenAI endpoint URL: https://.openai.azure.com/openai/v1/
✓ OpenAI endpoint: https://-openai.azure.com/openai/v1/
✓ OpenAI deployment: gpt-4o

▶ Configuring Entra ID
Create a NEW Entra ID app registration? (y/n) [y]:
Registering app: Fabric Storyboard Canilat (dev)
✓ App registered:
Setting Application ID URI...
Exposing access_as_user scope...
✓ Scope: api:// /access_as_user
Adding redirect URIs...
Adding API permissions...
Authorizing Office clients...
✓ Office clients authorized (SSO)
Generating client secret...
✓ Client secret created (valid 1 year)
Creating service principal...
Granting admin consent...
✓ Entra ID setup complete

▶ Installing dependencies
✓ Frontend packages installed
✓ Backend packages installed

▶ Building application
✓ Frontend → dist/
✓ Backend → api/dist/

▶ Running tests
✓ Frontend tests passed
✓ Backend tests passed

▶ Provisioning Azure infrastructure (Bicep via azd)
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13

To update to the latest version, run:
winget upgrade Microsoft.Azd
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13

To update to the latest version, run:
winget upgrade Microsoft.Azd
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13
```

One idempotent deploy: release and verify.

It deploys, verifies, and reports, every time the same way.

- Provisioned in ~2 minutes, then the app is deployed to Azure.
- Generates the production manifest, wires redirect URIs and RBAC (role-based access control).
- Ends with Deployment Complete and a summary of every resource and URL.
- The same script runs in the pipeline on every change, repeatable, not hand-made.

```

Cancelling stale deploy x PowerShell x + v
Subscription:
Location: East US 2

You can view detailed progress in the Azure Portal:
https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2F...

(✓) Done: Resource group: rg-fabricaddina (7.596s)
(✓) Done: Key Vault: (24.916s)
(✓) Done: Log Analytics workspace: log-appi (28.104s)
(✓) Done: Application Insights: appi-s (8.275s)
(✓) Done: Static Web App: swa:

SUCCESS: Your application was provisioned in Azure in 2 minutes 1 second.
You can view the resources created under the resource group
https://portal.azure.com/#@/resource/subscriptions,
WARNING: your version of azd is out of date, you have 1.23.12 and the latest stable version is 1.23.13

To update to the latest version, run:
azd get upgrade Microsoft.AzureStaticWebAppsCLI

✓ Resource
✓ Static Web
✓ Key Vault
✓ URL: https://

Deploying application to Azure

Welcome to Azure Static Web Apps CLI (2.0.8)

Deploying front-end files from folder:
C:\Users\frgisber\repo-addinfabric\OfficeAddin\dist

Deploying API from folder:
C:\Users\frgisber\repo-addinfabric\OfficeAddin\api

Consider providing api-language and version using --api-language and --api-version flags,
otherwise default values apiLanguage: node and apiVersion: 16 will apply

Deploying to environment: production

Deploying project to Azure Static Web Apps...
- Preparing deployment. Please wait...
✓ Deployed to https://.azurestaticapps.net

Generating production manifest
✓ dist/manifest-prod.xml

Adding production redirect URI to Entra app
✓ https://.azurestaticapps.net/dialog.html

Generating local development settings
✓ api/local.settings.json updated
✓ .env updated

Setting up RBAC for local dev (Cognitive Services OpenAI User)
⚠ Could not find OpenAI resource - assign role manually

[✓] Deployment Complete!

Application URL
Environment
Resource Group
Entra Client ID
Entra Client Secret
Tenant ID
Key Vault
Production Manifest
  
```


Real scenario.

Life imitates the demo: a story we did not plan

09





"I bought my car back."



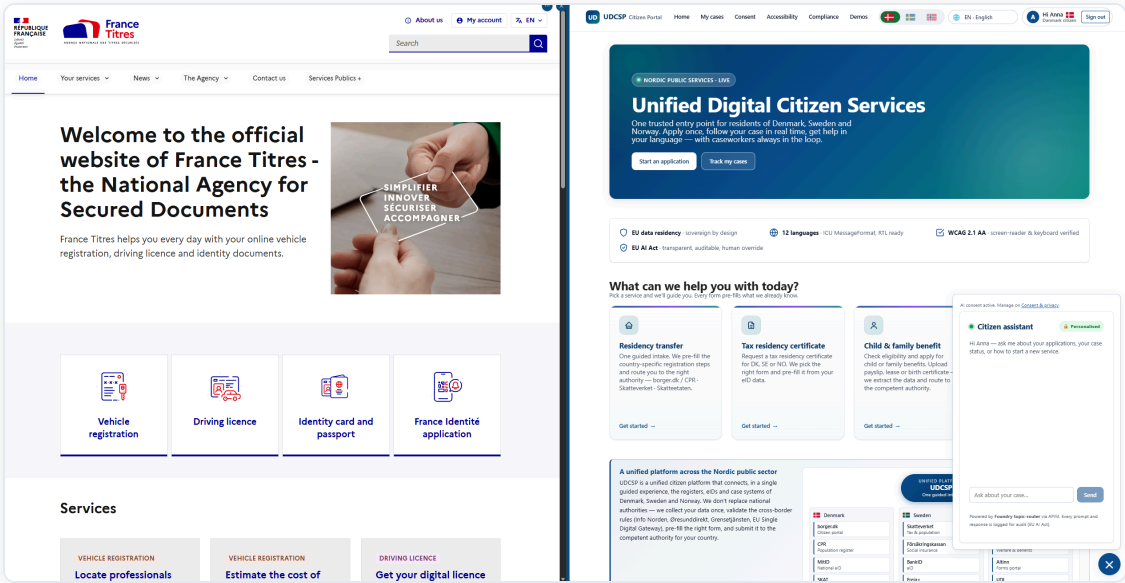
To change my licence plate, I signed in to the official French government portal: France Titres.

I had never seen it while building UDCSP.

And there it was: the same one-front-door layout, the same service cards (vehicle registration, driving licence, ID & passport), the same assistant in the corner.



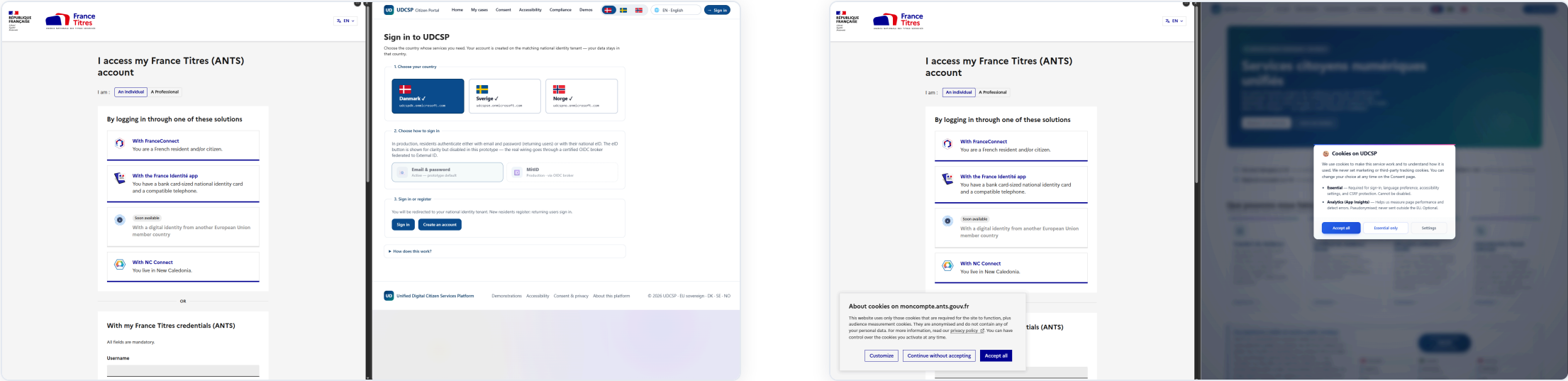
FR FRANCE TITRES, THE REAL THING



UDCSP, OUR DESIGN



...and we'd built it first.



Pick your country, sign in with a national eID, and ("coming soon") an EU cross-border identity. That is our model, almost line for line. We designed it before we ever saw it.



Status & roadmap.

What runs today · what's blueprint · how it reaches production

10





Requirement coverage: the honest scorecard.

Unified identity Entra External ID · eIDAS-ready ● Live ● Verified ID	12-language portal WCAG 2.1 AA accessibility ● Implemented	AI assistant web · mobile · voice ● Live	
Cross-border case DK → SE pre-fill + saga ● Live ● saga	Eligibility + human-in-loop verdict, then a person decides ● Implemented ● live TEE	EU AI Act evidence article-by-article trail ● Implemented ● live ledger	
GDPR rights records · access · erasure ● Scripted	Sovereign monitoring 9 workbooks, per country ● Live ● Fabric F64	One-command deploy 25 phases, idempotent ● Live	
● Live: today	● Implemented: merged & tested	● Scripted: idempotent installer	● Blueprint: designed & gated

The working core carries the demo; each blueprint brick has a named gate.

Four gates from accelerator to production.

GATE 1 · WK 1-2

Tenant validation

Confirm model deployments, Fabric F64 capacity, DDoS plan & Foundry hub.

GATE 2 · WK 3-6

Live confidential compute

Eligibility on a SEV-SNP enclave + Confidential Ledger.
Demo 6 → Live.

GATE 3 · WK 6-14

Partner-agency integration

Mutual-TLS agreements + production federation hub.
Demo 1 → real authority.

GATE 4 · WK 14-20

D365 + Copilot for Service

Strangler-fig repoint of case writes. **Demo 5 → Live.**

Today → Gate 1 → Gate 2 → Gate 3 → Gate 4 → production. Each gate is independently shippable: every blueprint brick has an owner, a window, and a demo it flips to live.

Demo.

10 minutes, live on a real tenant



The run of show.

01 Anna moves DK → SE ● Live

Sign in with a Danish eID · upload a passport and lease · the AI reads, translates and proposes a result · consent · the case crosses the border.

02 Lars calls the voice line ★ ● Live

A blind citizen dials a real free number · the model answers in Norwegian and routes itself · a human transfer is offered.

03 Maria uses a screen reader in Polish ● Live

The whole portal in Polish, keyboard only, screen-reader clean: accessibility is a citizen right.

04 Erik photographs a payslip on iPhone ● Live

The same web app on mobile · native iOS picker · structured fields and a result, inline.

05 Ole builds it from a clean tenant ● Live

One command: 25 phases, sample data seeded, smoke tests green.

Hero moment: a real phone call: dial `+33 801 150 799`, hear the model answer, and watch the live dashboard update within two minutes.



One command, from clean tenant to running platform.

```
git clone https://github.com/fredgis/UDCSP
cd UDCSP
pwsh ./scripts/install/Install-UDCSP.ps1 `
    -Environment dev -Zone all `
    -SeedSyntheticData -Verbose
```

- 25 phases in order: landing zone → identity → security → data → monitoring → gateway → Foundry → Dataverse → frontend → voice → governance → quality
- Sample data is seeded in parallel with the frontend
- A smoke test runs at the end (identity · gateway · AI · case creation · dashboard · accessibility)
- An HTML report is saved in `scripts/install/reports/<timestamp>/`

From `git clone` to a working federated platform with real sample data: one command. The same script runs in our pipeline on every pull request.

CLOSING

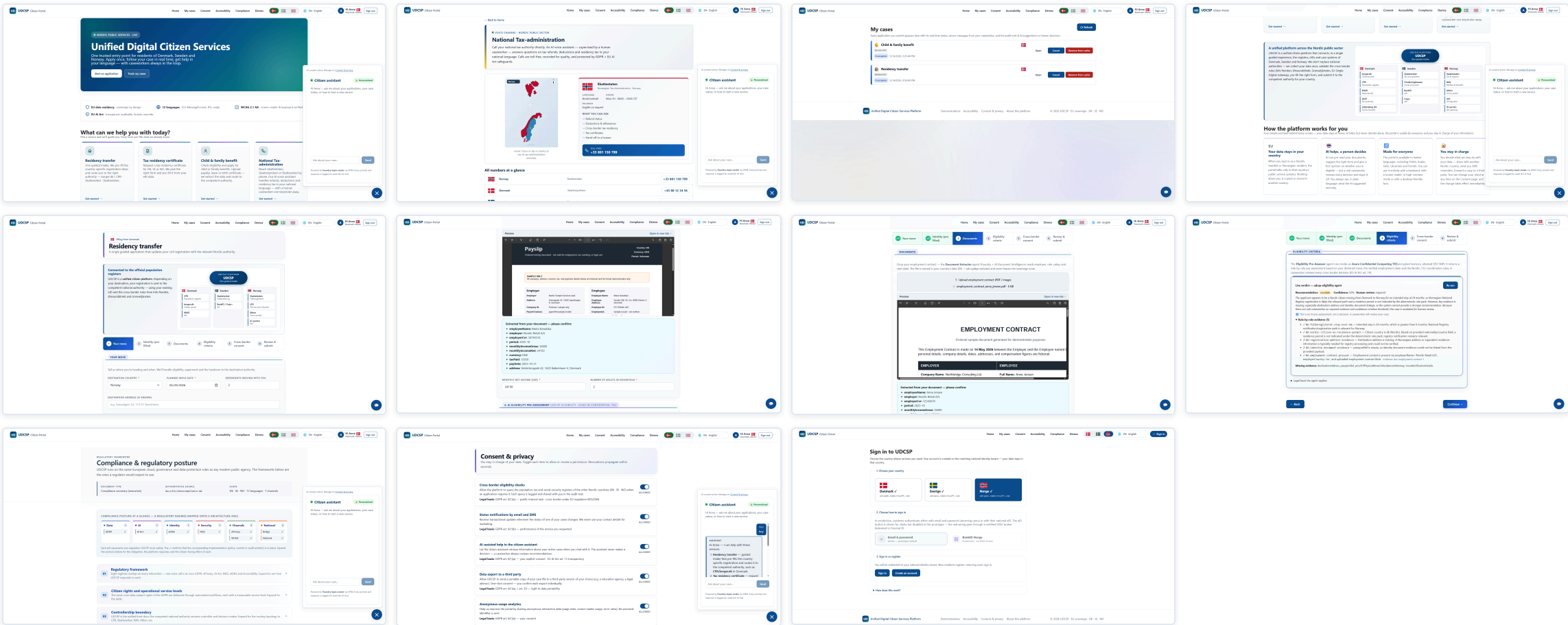
UDCSP: a production accelerator with a named path to production.

3 sovereign zones · 7 AI agents · one private path from the portal to the case · every decision anchored to a regulation · every claim labelled and provable on a real Azure tenant. A working core you deploy with one command today, and a four-gate roadmap that takes the blueprint bricks to production. A real French government portal, seen by accident, proved the architecture was right.

Frederic Gisbert · August 2026 · github.com/fredgis/UDCSP



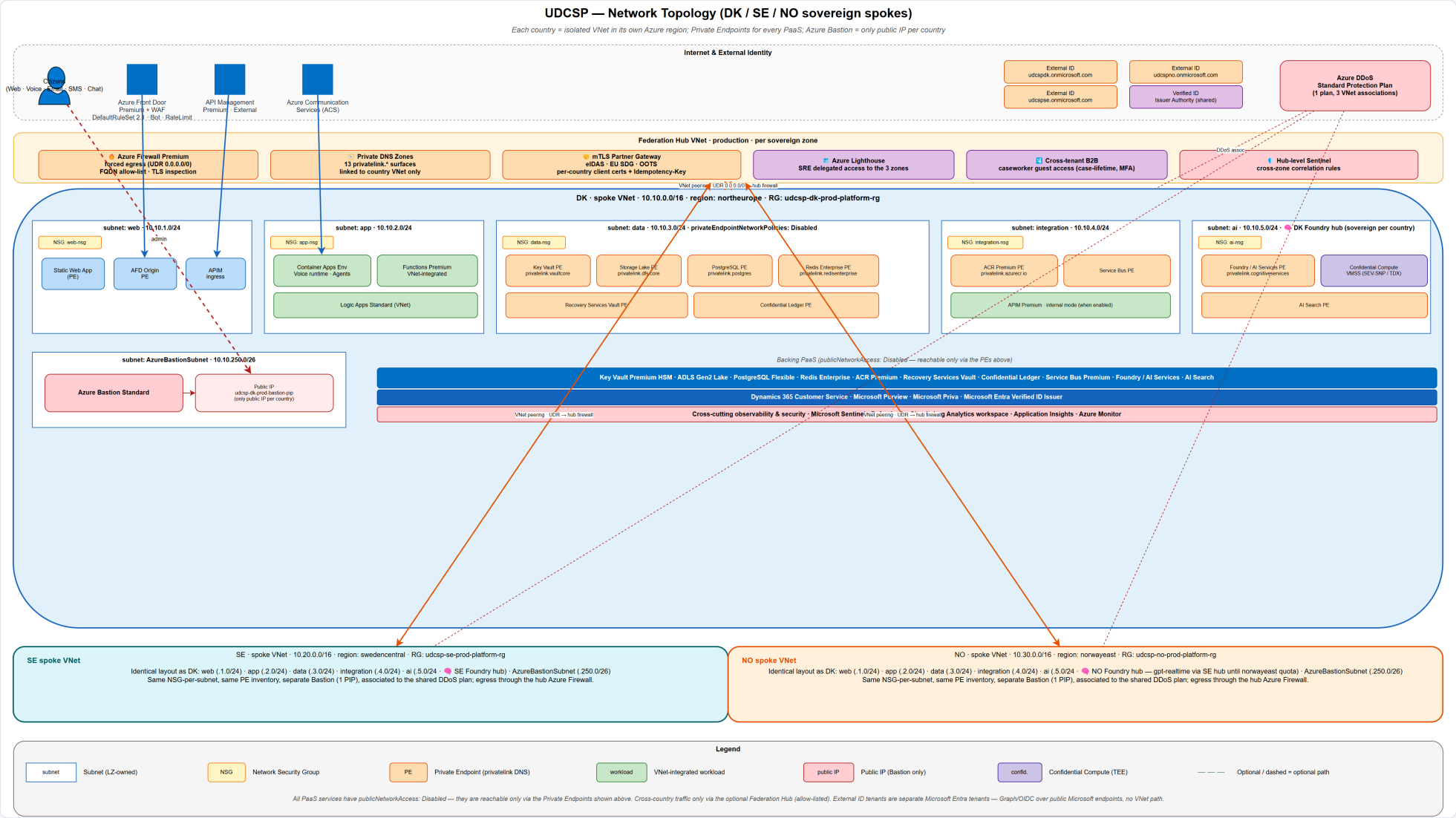
Annex: the platform, screen by screen.



1 Home · 2 Contact / voice · 3 My cases · 4 Assistant · 5 Residency intake · 6-7 Document reading · 8 Cross-border result · 9 Compliance · 10 Consent · 11 Sign-in. One web app, twelve languages, three channels, using the same React and TypeScript code on desktop, tablet and phone.

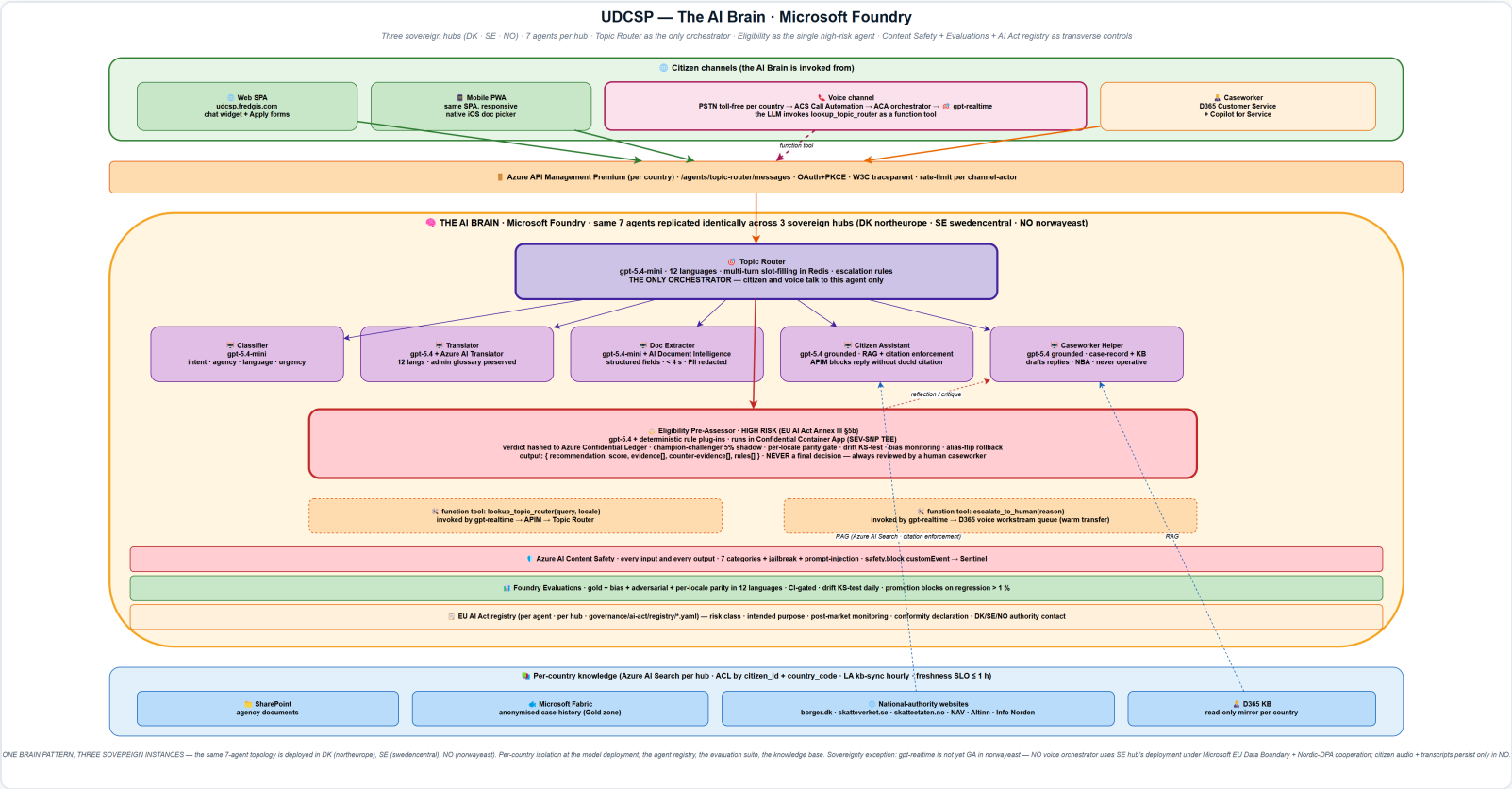


Annex: network topology in full.





Annex: the AI Brain blueprint in full.



One orchestrator and six specialists on Azure AI Foundry, the high-risk eligibility agent in a sealed enclave, and the EU AI Act register wired into every model call: the dense source map behind the simplified gateway slide.

Annex: telemetry pillars in full.

METRICS / LOGS

Azure Monitor

3 workspaces, one per country · 180 days hot · 7 years cold · never joined

DISTRIBUTED TRACE

Application Insights

One trace number end-to-end · front door → gateway → workflow → Dataverse → agent → model

AI QUALITY

Foundry Evaluations

Continuous · drift monitors · language parity · bias monitoring

AI ACT EVIDENCE

Confidential Ledger

Records that cannot be changed · joins logs, AI and Dataverse in one query

ACTIVE

Synthetic + real-user

5 external regions · 60-second probes · load times per language

DASHBOARDS

9 workbooks

platform health · citizen-journey funnel · AI-decision traces · per country